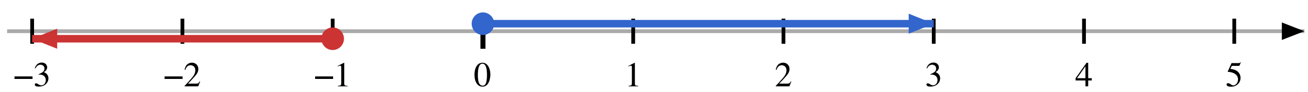
Lecture 7: Vectors

One dimensional vectors

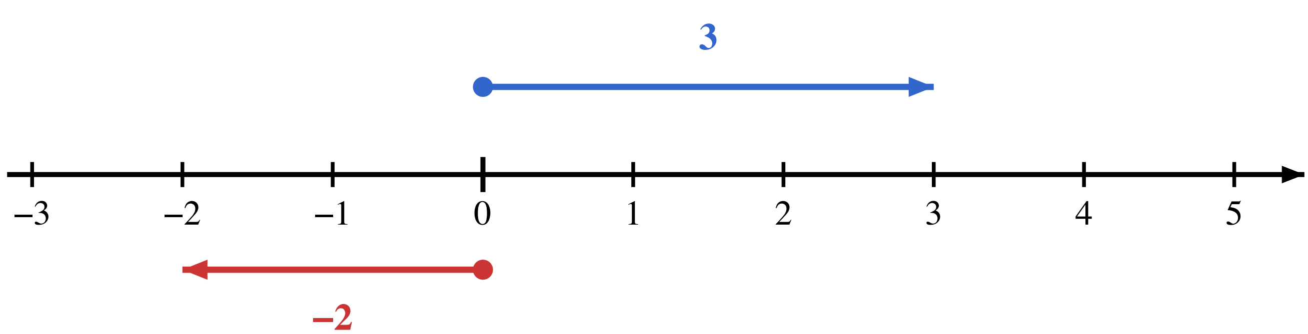
A vector is a directed segment. It has both magnitude and direction. One dimensional vectors sit on a number line. The direction is pointing to either left or right. Examples are as below for two vectors on the line:



The starting point of the vector is called the **tail**, and the ending point is called the **head**. The length of the vector is called its **magnitude**. The direction is indicated by the position of the head relative to the tail. If the head is to the right of the tail, the vector points to the right; if the head is to the left of the tail, the vector points to the left.

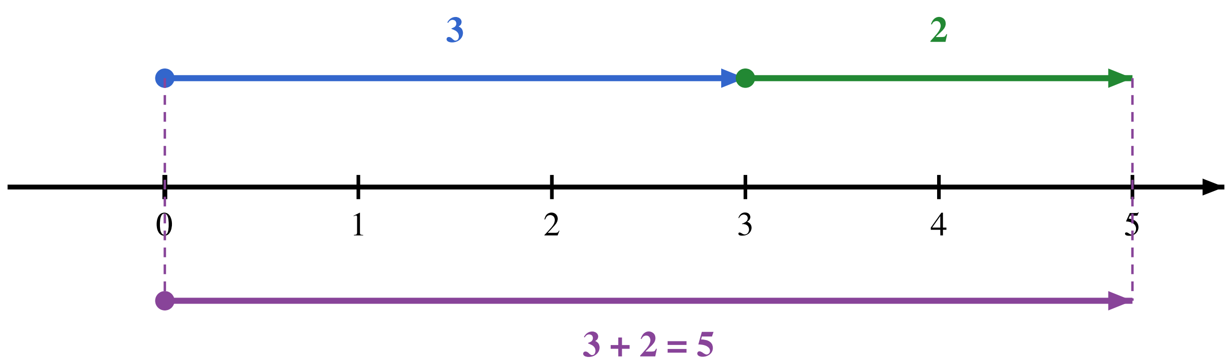
Standard representation

If the beginning point sits at zero, then the vector can be represented simply by its endpoint value. For example, the vector that starts at 0 and ends at 3 can be represented as $\overrightarrow{3}$, and the vector that starts at 0 and ends at -2 can be represented as $\overrightarrow{-2}$. The direction is indicated by the sign of the number.



Addition and subtraction of one dimensional vectors

Vectors can be **added** together by placing them head to tail. For example, adding the vectors $\overrightarrow{3}$ and $\overrightarrow{2}$ results in a vector that starts at 0 and ends at 5:

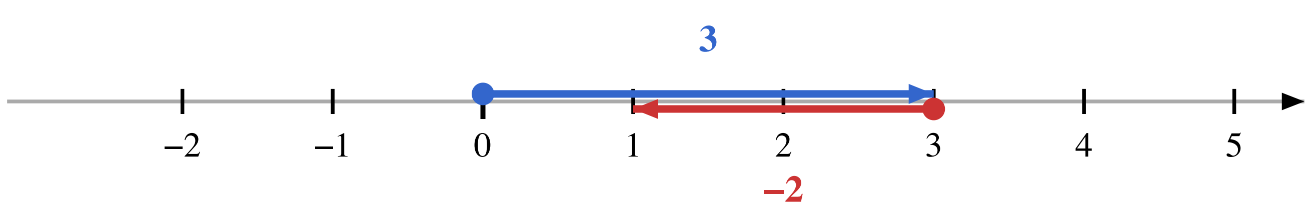


In previous lectures, we know we can move a segment to any point on the number line without changing its length by constructing parallel lines and parallelograms. So we can move the vector $\overrightarrow{2}$ to start at the head of the vector $\overrightarrow{3}$ without changing its length. This means that adding two vectors is equivalent to adding their magnitudes.

So if we have two numbers a and b representing two vectors $\overrightarrow{a}$ and $\overrightarrow{b}$, then their sum is given by:

$$\overrightarrow{a} + \overrightarrow{b} = \overrightarrow{a + b}$$

How about subtraction? Subtraction can be thought of as adding a negative vector. For example, subtracting the vector $\vec{2}$ from the vector $\vec{3}$ can be visualized as adding the vector $\overrightarrow{-2}$ to the vector $\overrightarrow{3}$:

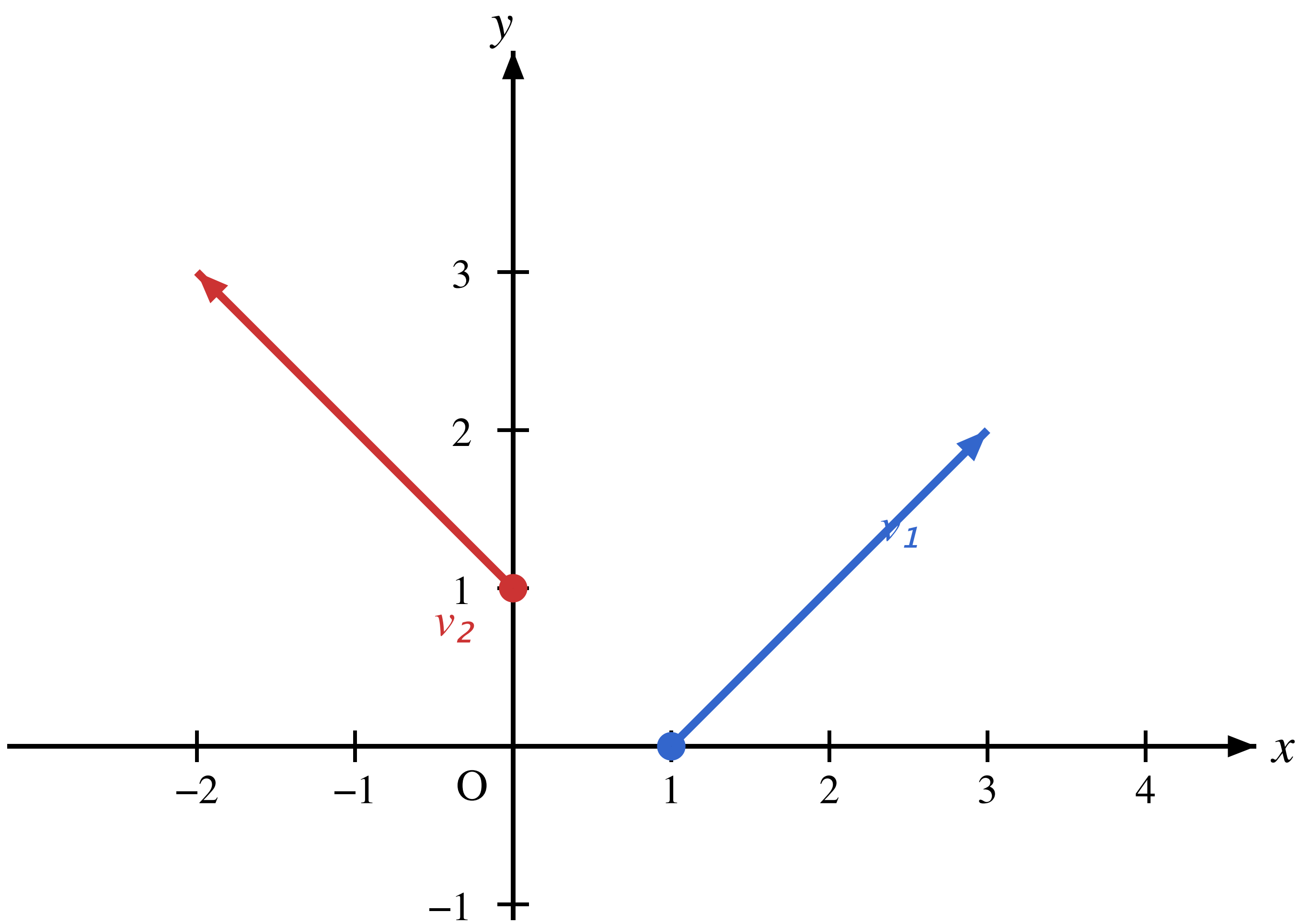


So if we have two numbers a and b representing two vectors $\overrightarrow{a}$ and $\overrightarrow{b}$, then their difference is given by:

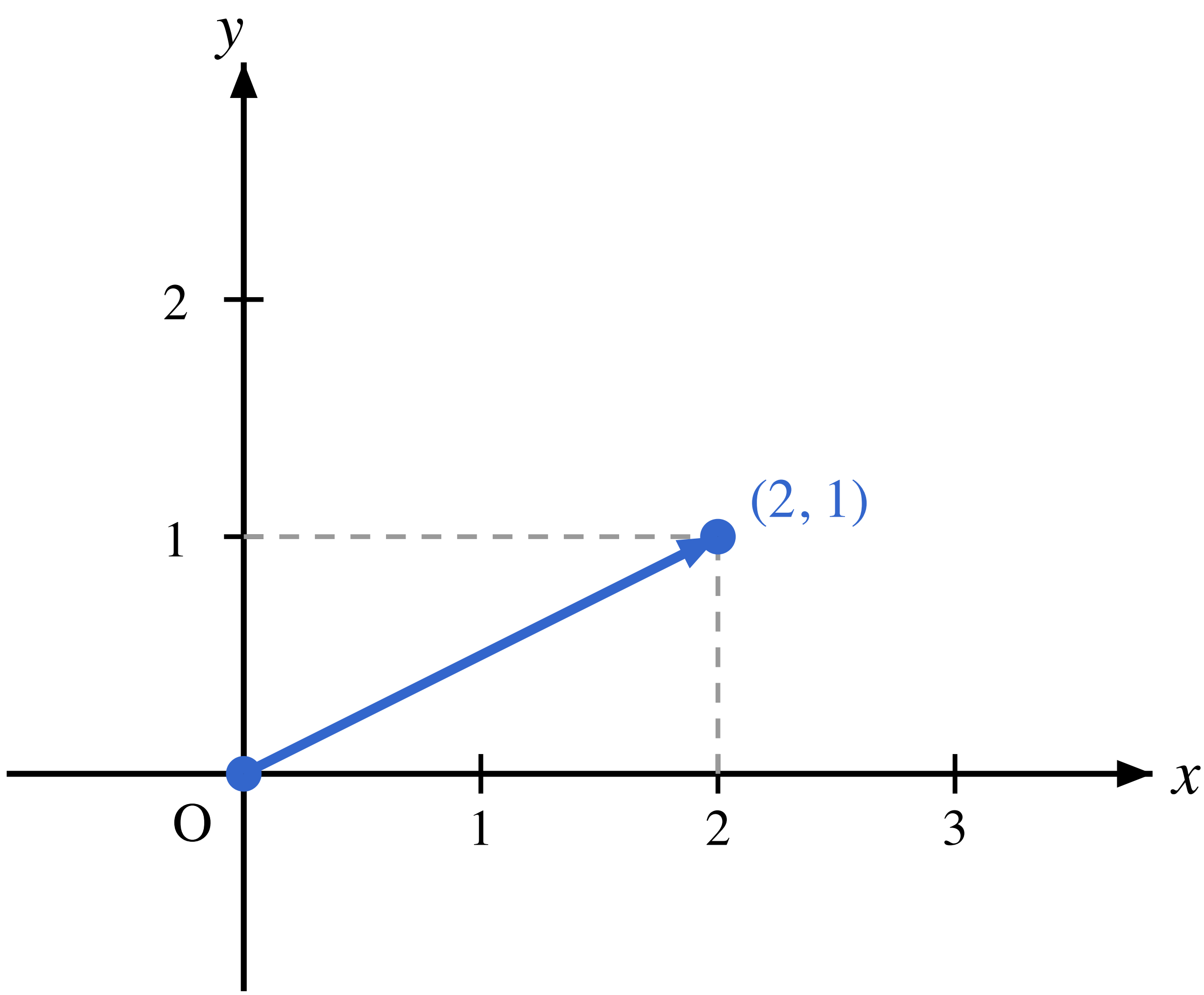
$$\overrightarrow{a} - \overrightarrow{b} = \overrightarrow{a} + \overrightarrow{-b} = \overrightarrow{a - b}$$

Two dimensional vectors

Two dimensional vectors can be represented as directed segments in the plane. See following two vectors in the plane:



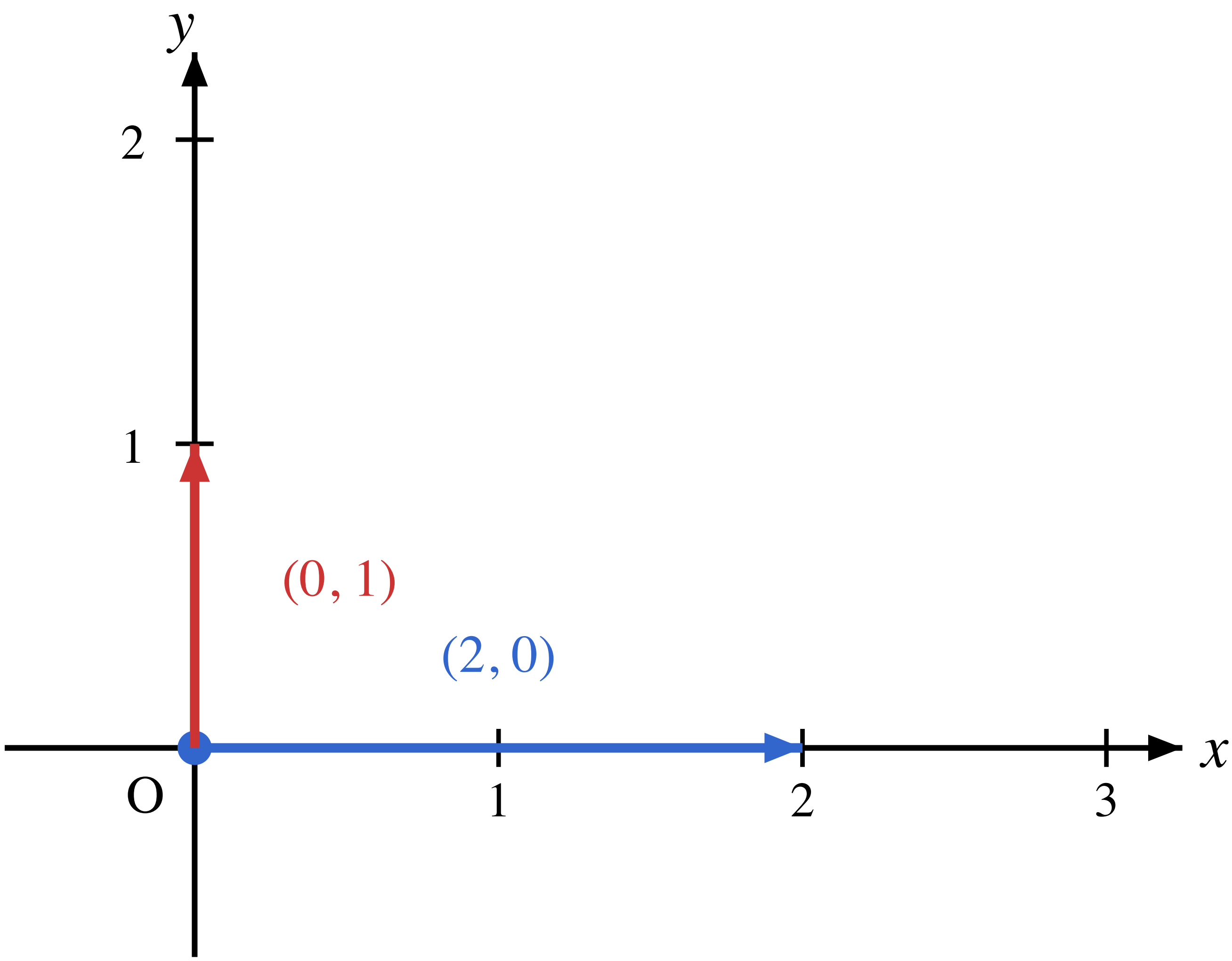
If the head of the vector sits at the origin $(0, 0)$, a two dimensional vector can be represented by an ordered pair of numbers (x, y), where x is the horizontal component and y is the vertical component. For example, the vector that starts at the origin $(0, 0)$ and ends at the point $(2, 1)$ can be represented as $\overrightarrow{(2, 1)}$.



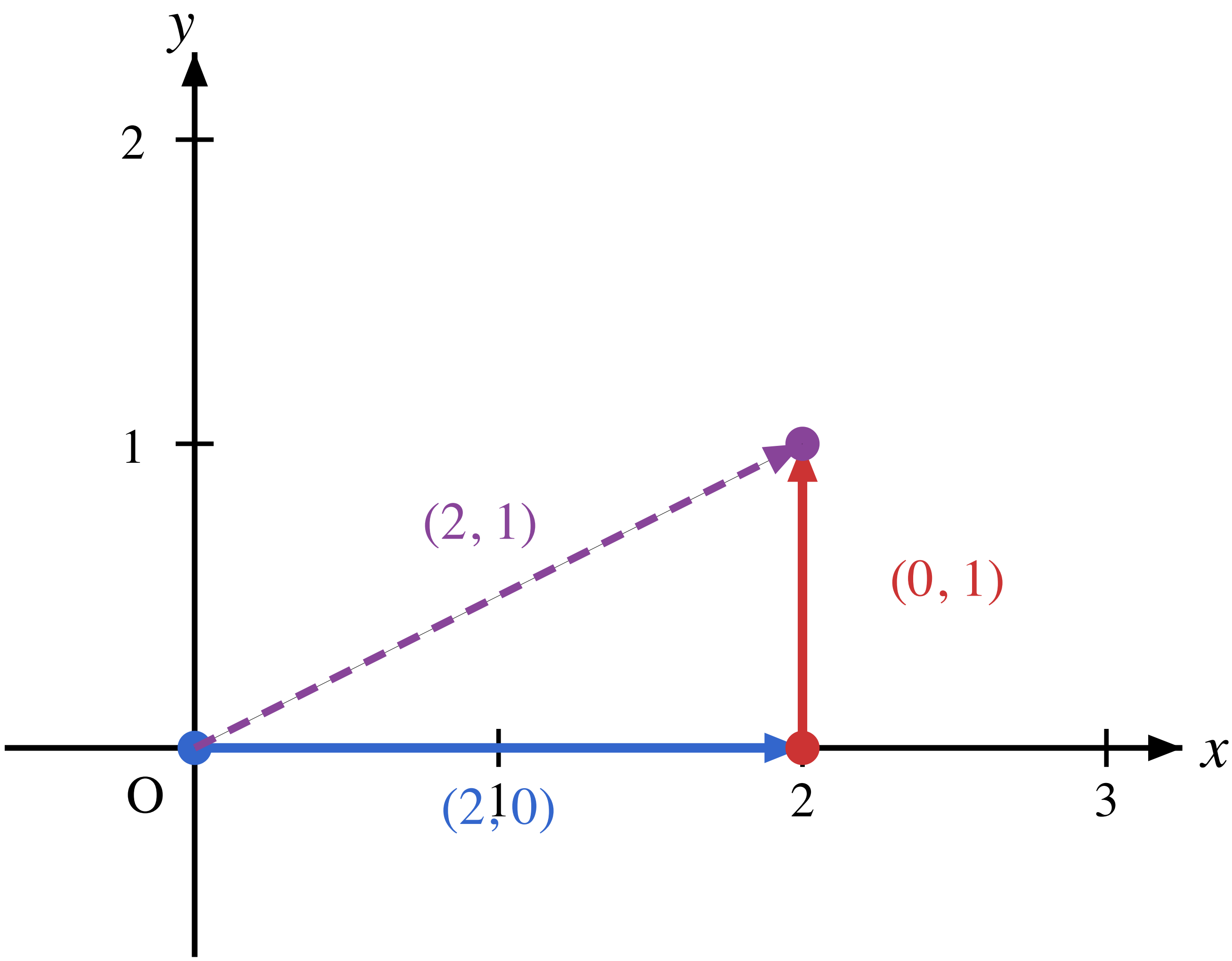
Addition of two dimensional vectors

How about the addition of two dimensional vectors? Let us consider a special case first. We want to consider the addition

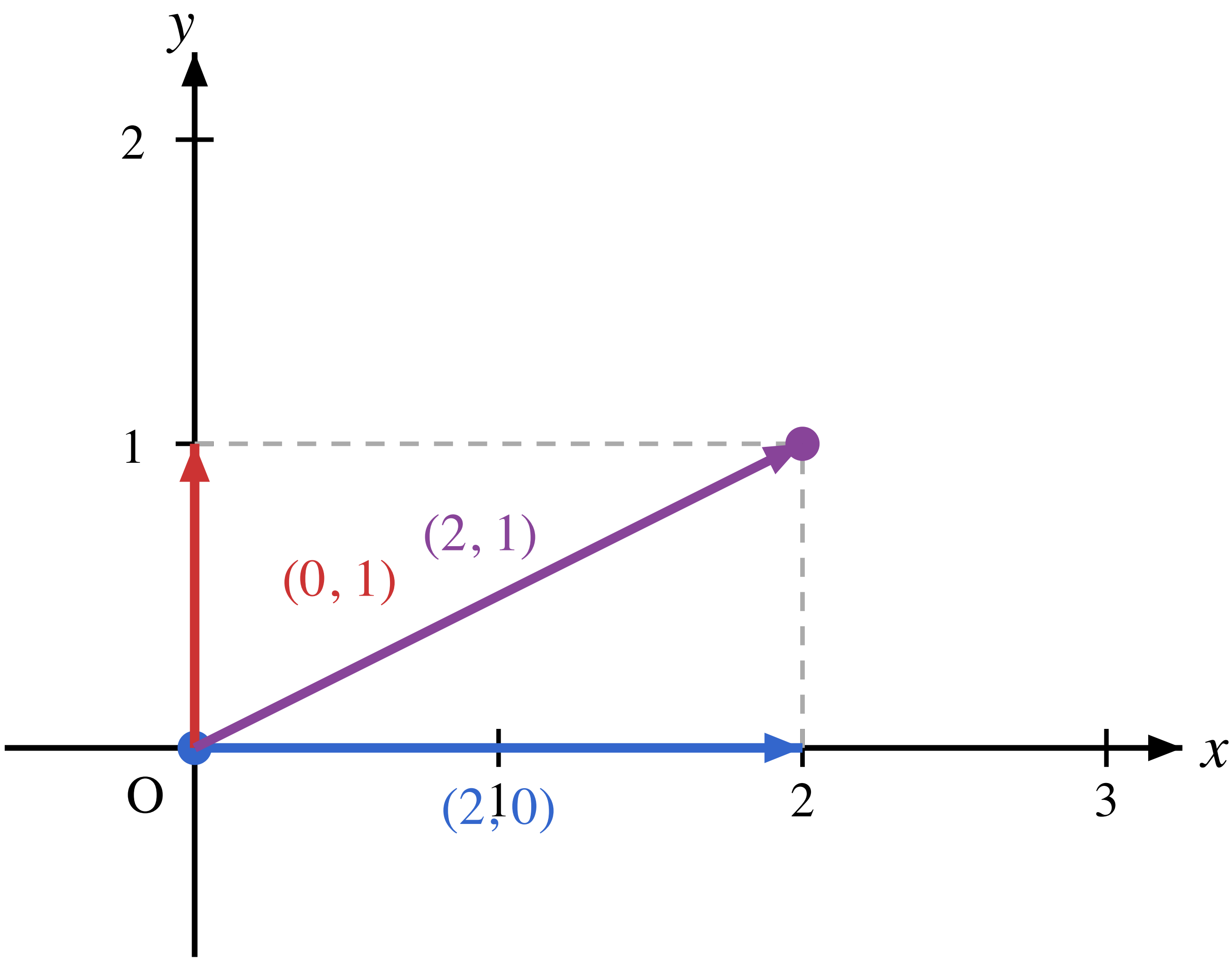
$$\overrightarrow{(2,0)} + \overrightarrow{(0,1)}$$



First geometric interpretation, the vector $\overrightarrow{(2,0)}$ means moving 2 units to the right, and the vector $\overrightarrow{(0,1)}$ means moving 1 unit up. So adding these two vectors results in a vector that starts at the origin and ends at the point $(2,1)$. This is the **head-to-tail** method of vector addition generalizing the dimension one case.



Or equivalently, we can construct a parallelogram with the two vectors as adjacent sides. The resulting diagonal vector starting at the origin and ending at the point $(2, 1)$ is the sum of the two vectors.



In this special case, we can see that adding the two vectors results in a vector whose components are the sums of the corresponding components of the two vectors:

$$\overrightarrow{(2,0)} + \overrightarrow{(0,1)} = \overrightarrow{(2+0, 0+1)} = \overrightarrow{(2,1)}$$

In general, we have the following **canonical decomposition** of any two dimensional vector:

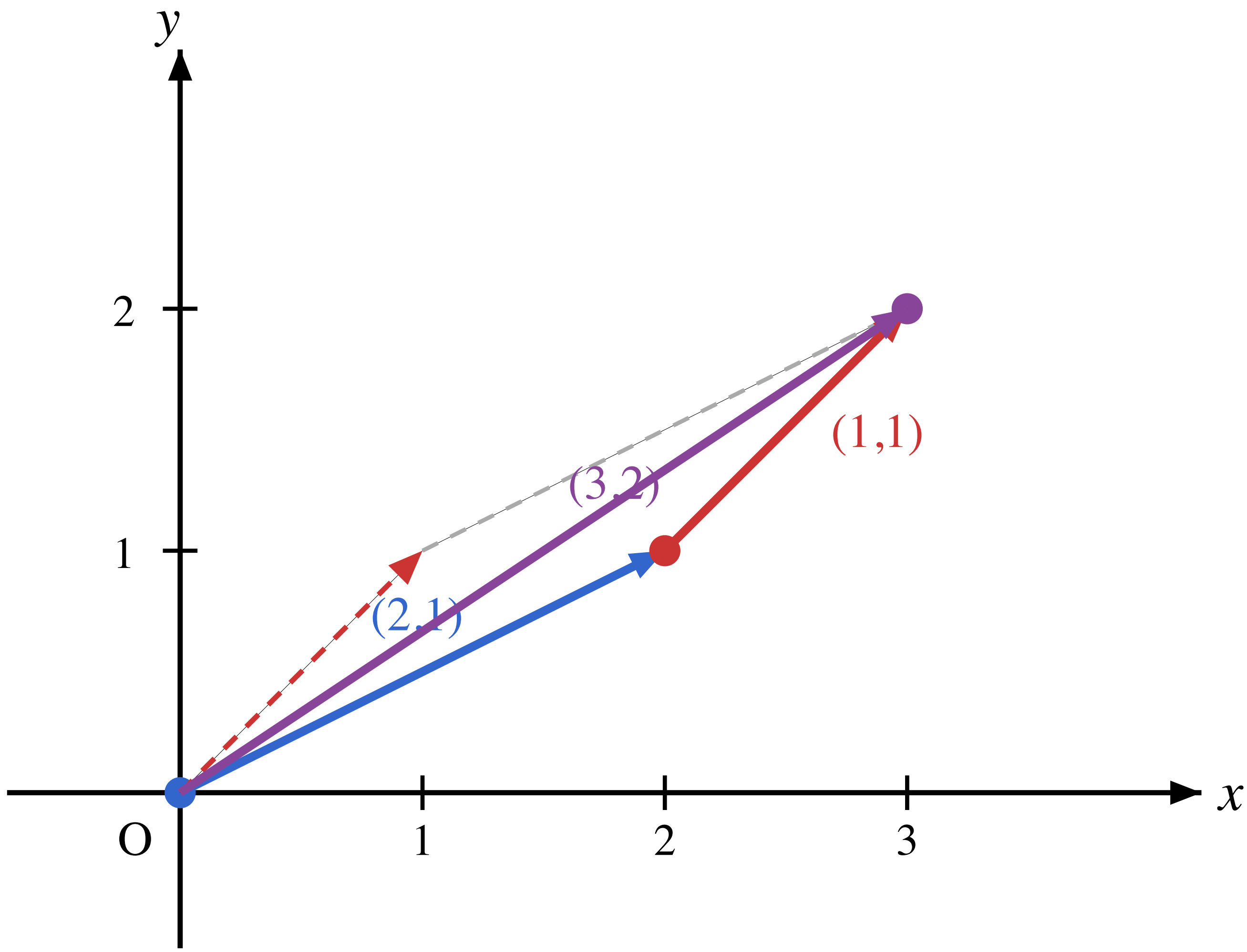
$$\overrightarrow{(x,y)} = \overrightarrow{(x,0)} + \overrightarrow{(0,y)}$$

We can view any vector $\overrightarrow{(x,y)}$ as moving x units in the horizontal direction and y units in the vertical direction.

Now for the general case of adding two vectors

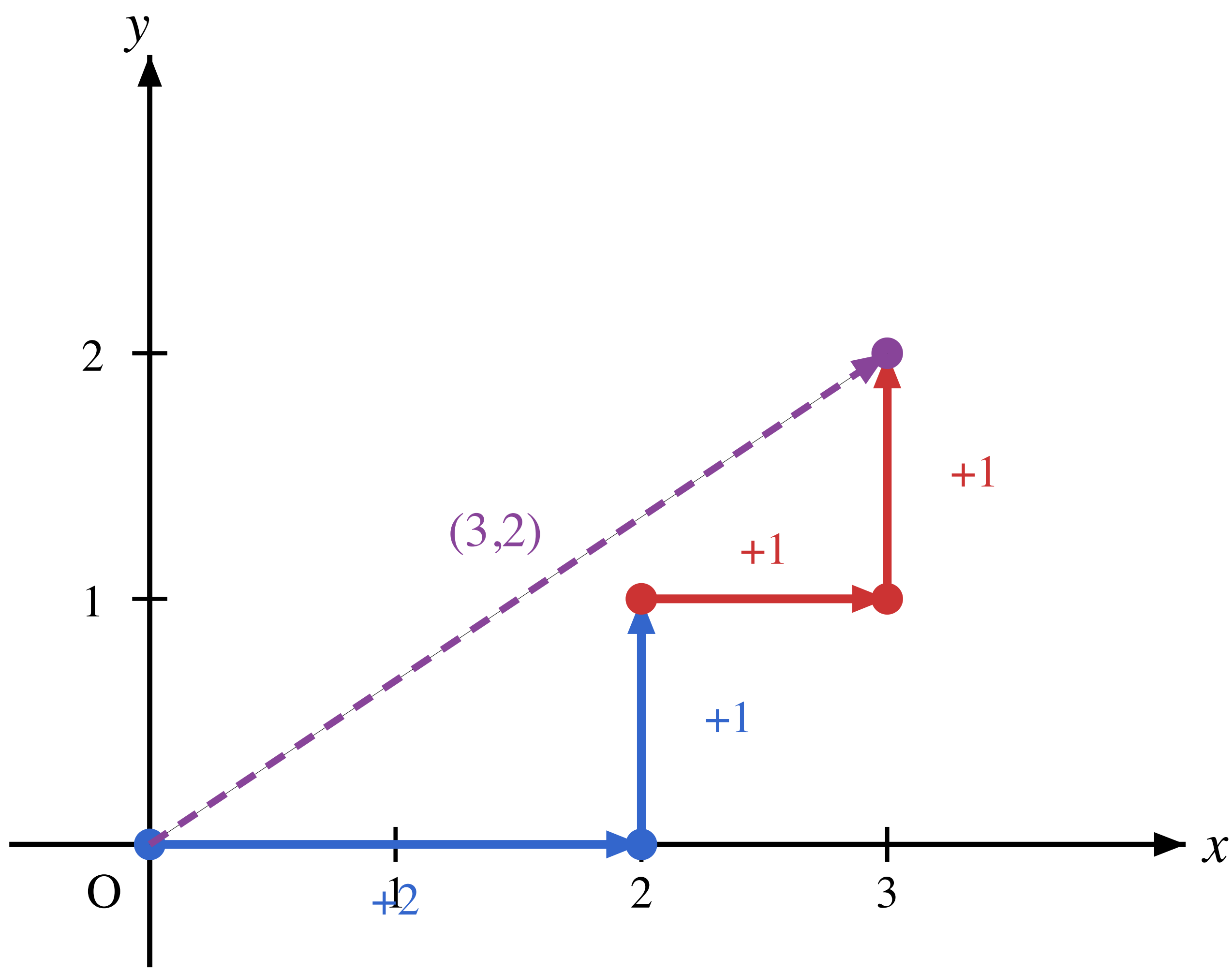
$$\overrightarrow{(2,1)} + \overrightarrow{(1,1)}$$

we can use the head-to-tail method or the parallelogram method to visualize the addition:



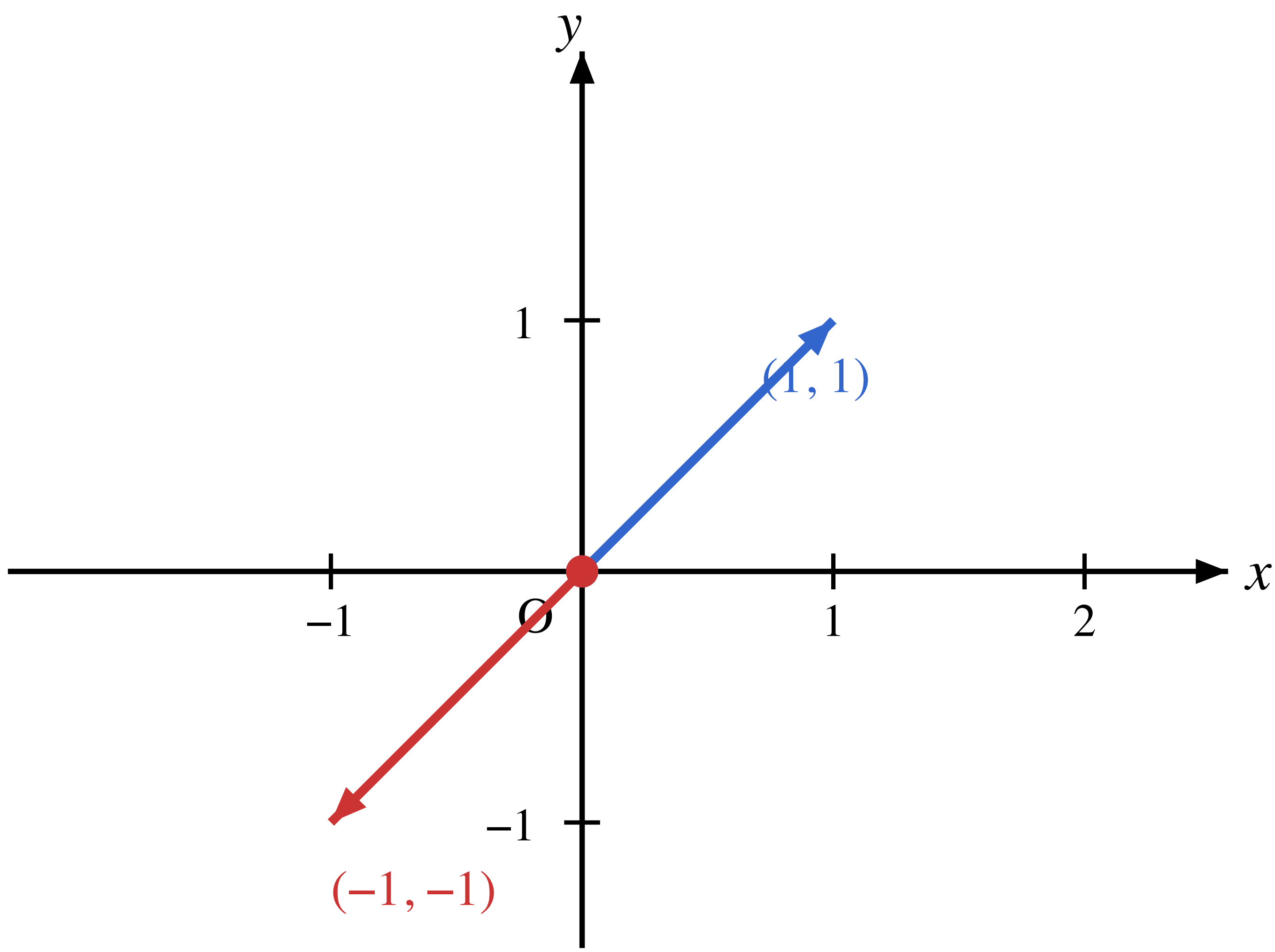
Actually, we can decompose both vectors into their horizontal and vertical components:

$$\overrightarrow{(2,1)} + \overrightarrow{(1,1)} = \left(\overrightarrow{(2,0)} + \overrightarrow{(0,1)}\right) + \left(\overrightarrow{(1,0)} + \overrightarrow{(0,1)}\right) = \overrightarrow{(3,0)} + \overrightarrow{(0,2)} = \overrightarrow{(3,2)}$$



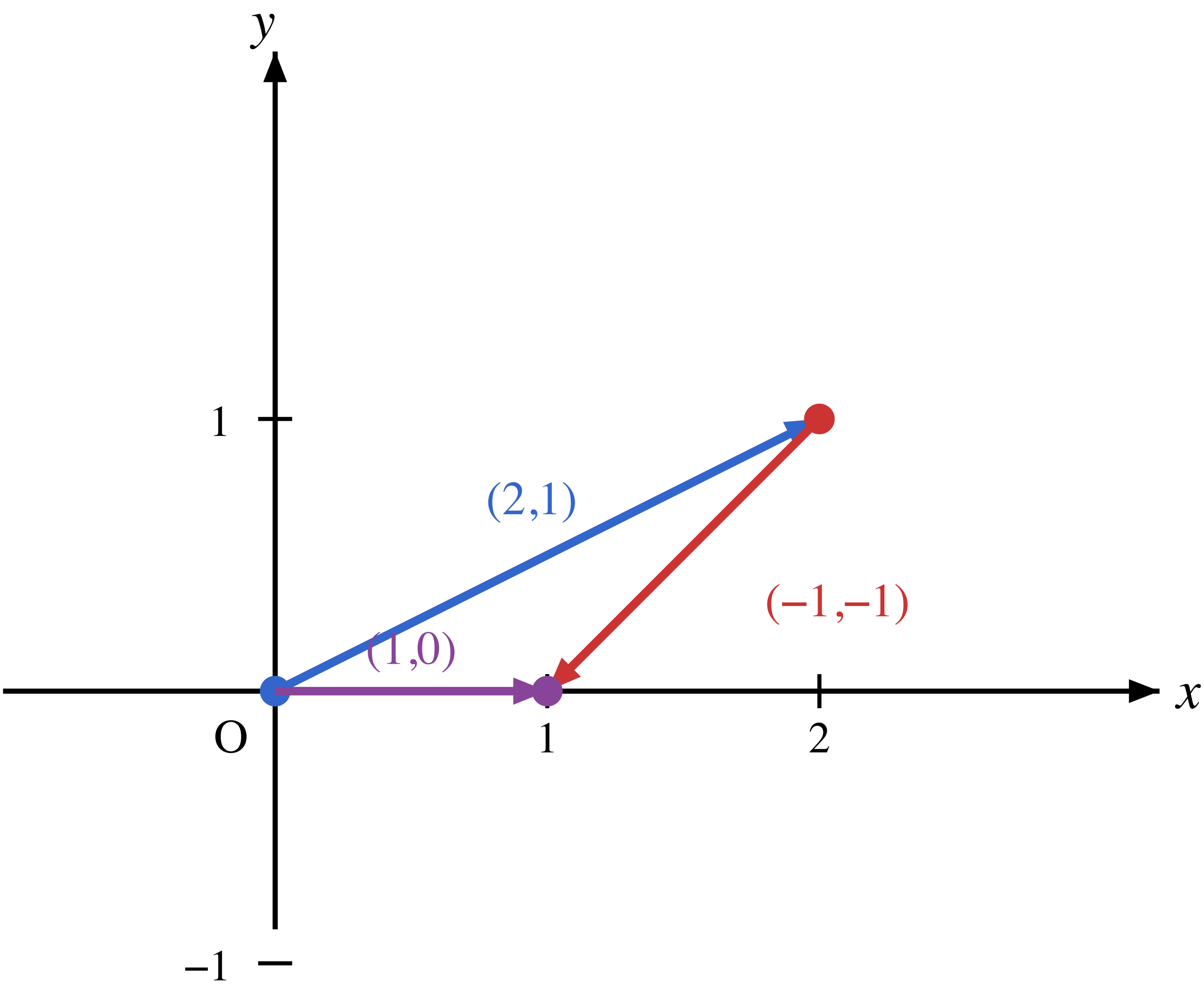
Subtraction of two dimensional vectors

First we define the negative of a vector. The negative of a vector $\overrightarrow{(x,y)}$ is the vector $\overrightarrow{(-x,-y)}$. Geometrically, this is the vector that has the same magnitude as $\overrightarrow{(x,y)}$ but points in the opposite direction.



Now we can define subtraction of two vectors as adding the negative of the second vector. For example, to subtract the vector $\overrightarrow{(1,1)}$ from the vector $\overrightarrow{(2,1)}$, we can add the negative of $\overrightarrow{(1,1)}$:

$$\overrightarrow{(2,1)} - \overrightarrow{(1,1)} = \overrightarrow{(2,1)} + \overrightarrow{(-1,-1)} = \overrightarrow{(1,0)}$$



The general formula for subtracting two vectors is:

$$\overrightarrow{(x_1, y_1)} - \overrightarrow{(x_2, y_2)} = \overrightarrow{(x_1, y_1)} + \overrightarrow{(-x_2, -y_2)} = \overrightarrow{(x_1 - x_2, y_1 - y_2)}$$